

SIGNmakerTM wayfinding

A new and original approach to signage design and fabrication



Game Changer

What is it?

It's a complete, automated, end-to-end process for the design and fabrication of visual and tactile wayfinding signs.

Who benefits?

Every organization that employs visual and tactile ADA wayfinding signage to help users navigate a system - rail transit, airports, hospitals, campuses, civic and cultural institutions.

Why is it important?

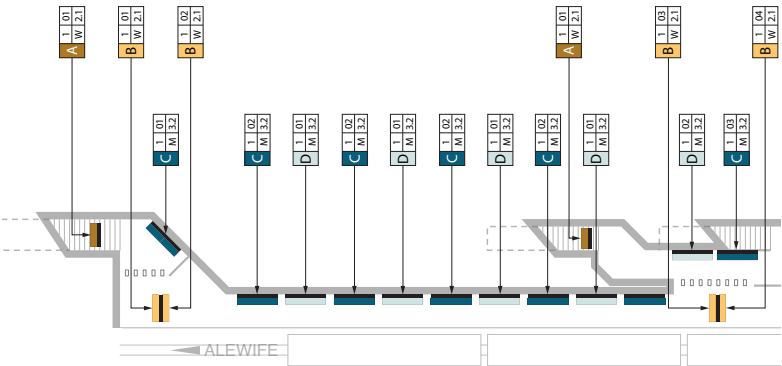
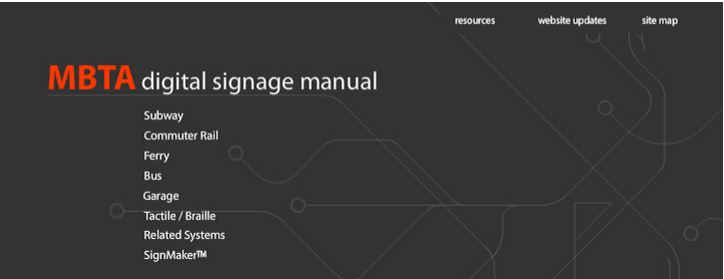
Because it saves time and money while vastly improving the quality and consistency of output. It is custom-designed for large systems.

Is it tested?

The MBTA has embarked on the process of replacing all visual and tactile signage for their entire system, including subway, commuter rail, ferry and bus.

Innovations

New System



Digital Signage Manual (DSM)

Server-based repository which houses all existing conditions information and design tools related to system signage.

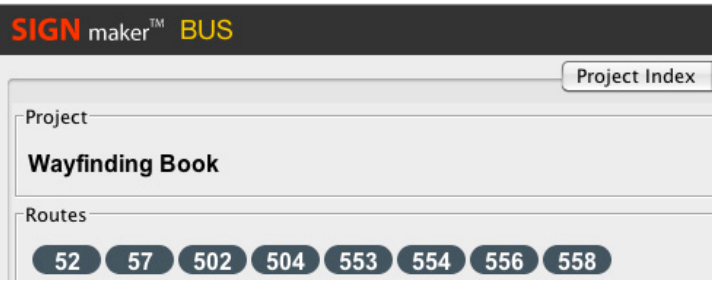
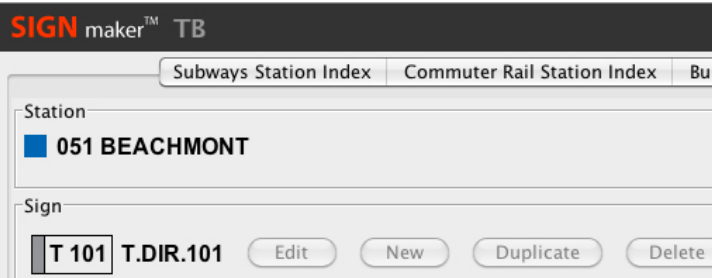
Data-Driven Graphic System

The entire signage system is built on individual station data and a newly-designed library of standard wayfinding graphic elements.

End-to-End Delivery Process

A fully integrated design, construction document, and fabrication process that enables in-house staff to control every aspect of signage layout, design, and delivery.

BIA Custom-Designed Software



A custom, menu-driven application for automating the design process and ensuring that visual wayfinding signs are correct, clear, and consistent across the system.



Enables automatic generation of full size, ADA-compliant Tactile/Braille signs. Output files are used for direct fabrication of etched zinc or stainless steel signs.

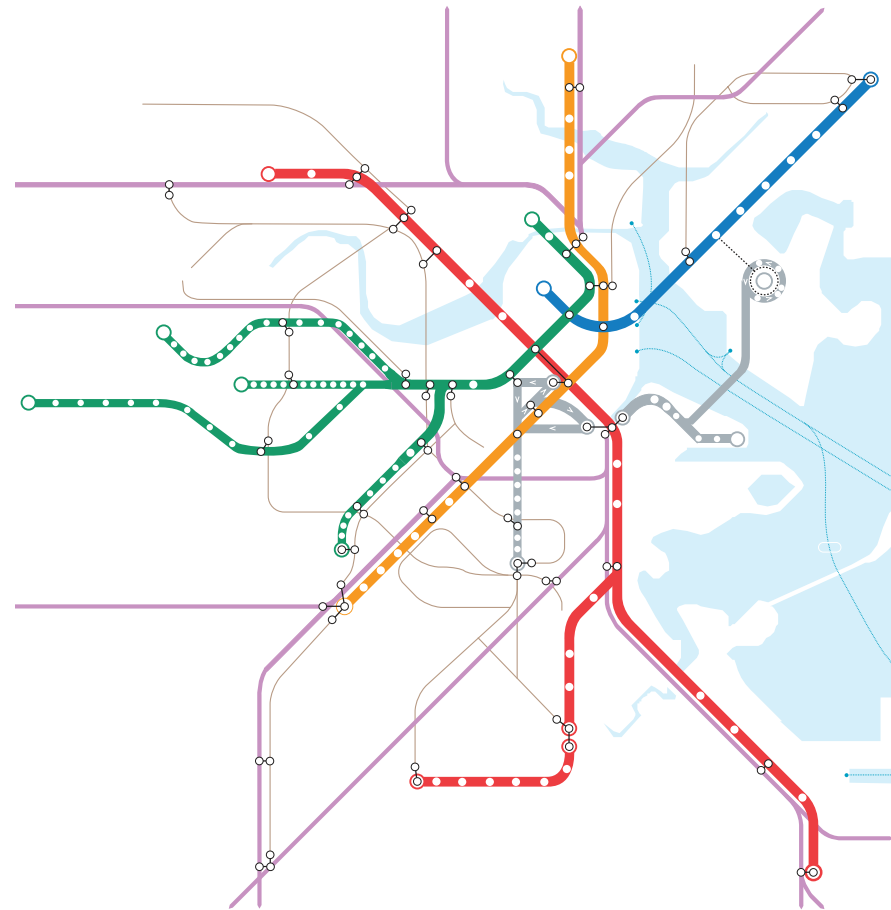


Enables automatic generation of full-size bus stop signs on standard templates. The software is linked to the MBTA's route database of 18,000 stops and serves as a content management tool for the MBTA's in-house Bus Operation staff.

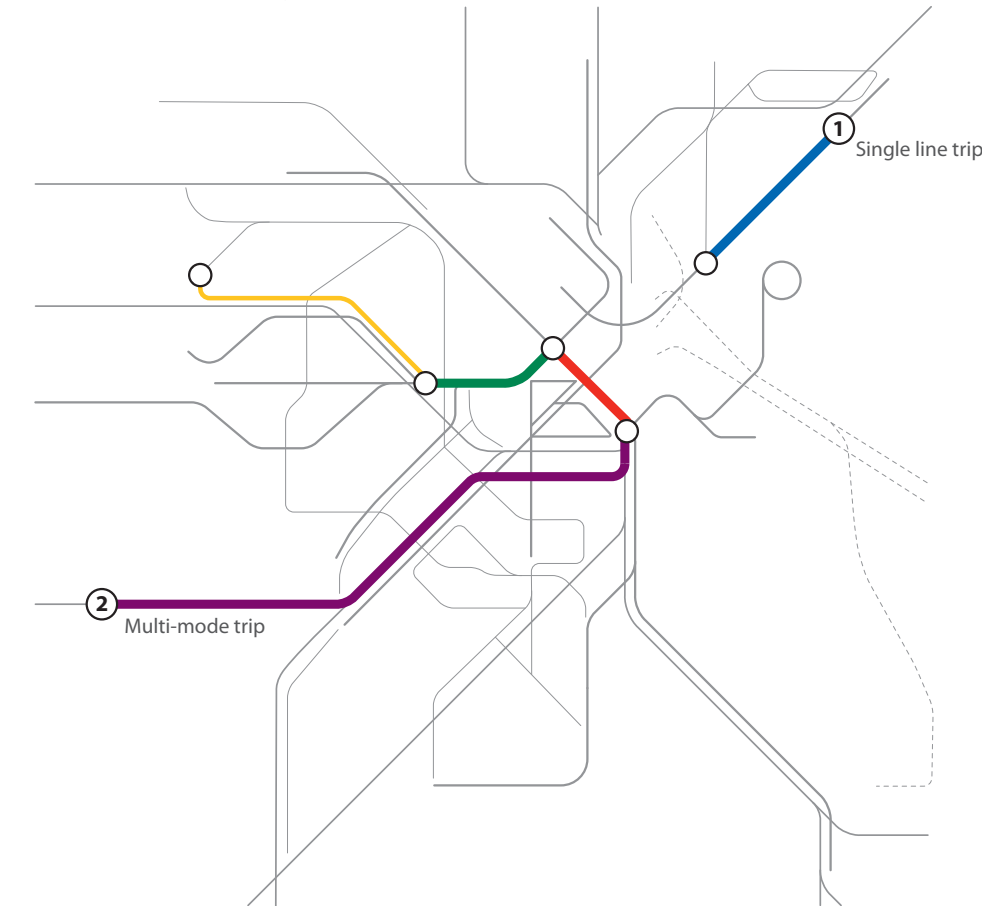
Creating “One System”

Bertaux+Iwerks Architects have completed a comprehensive redesign of the MBTA’s entire wayfinding system in order to create a unified signage system that links all modes of transit:

- 150 subway stations
- 130 train stations
- 8 ferry stations
- 18,000+ bus stops
- 40,000+ signs
- all automated



Customer Journey



All transit modes interconnect. A unified wayfinding system helps customers navigate from place to place, while helping the MBTA to project brand consistency and reliability.

Subway



Trolley



Train



Bus



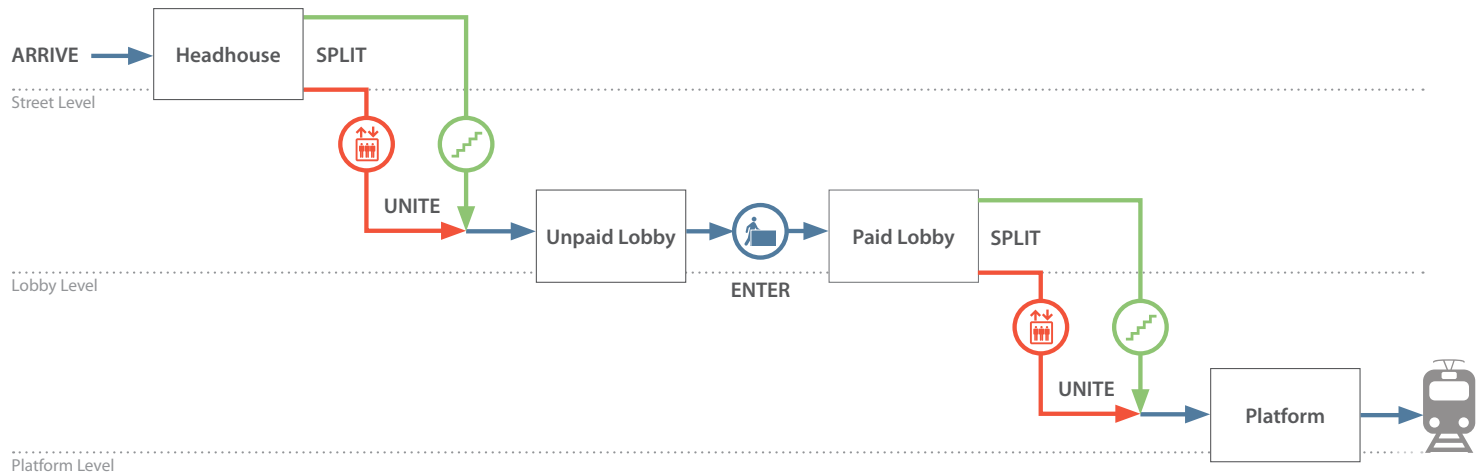
Ferry



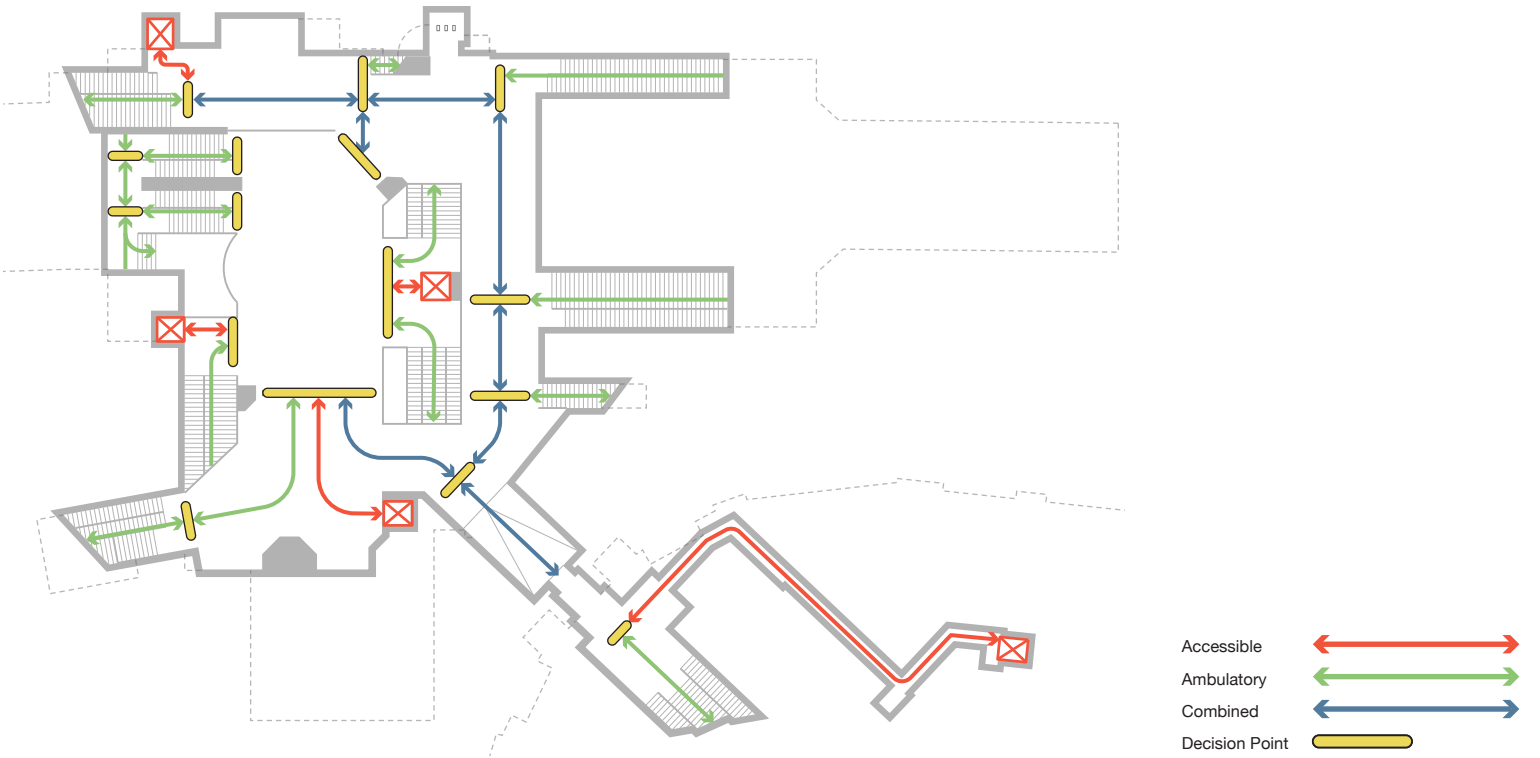
Understanding Customer Trips

“Create a consistent system of signage and location of signs throughout the MBTA system that gives users the ability to navigate the customer journey at each station.”

- BCIL/MBTA Settlement Agreement Outcomes



Station Paths



Supporting a diverse community of riders

While the majority of MBTA transit stations are accessible, the accessible paths connecting entrances to trains are frequently different than the ambulatory paths. The new wayfinding system is based on a pathway diagram system that identifies every path split and decision point within a station. Accurate signing of accessible, ambulatory and combined paths is determined by each station’s unique set of pathways.



Combined Path



Accessible Path



Ambulatory Path

Taking Inventory


Subway


Trolley


RED LINE


Subway


ORANGE LINE


Trolley


GREEN LINE


Subway


BLUE LINE


BRT


SILVER LINE


Train

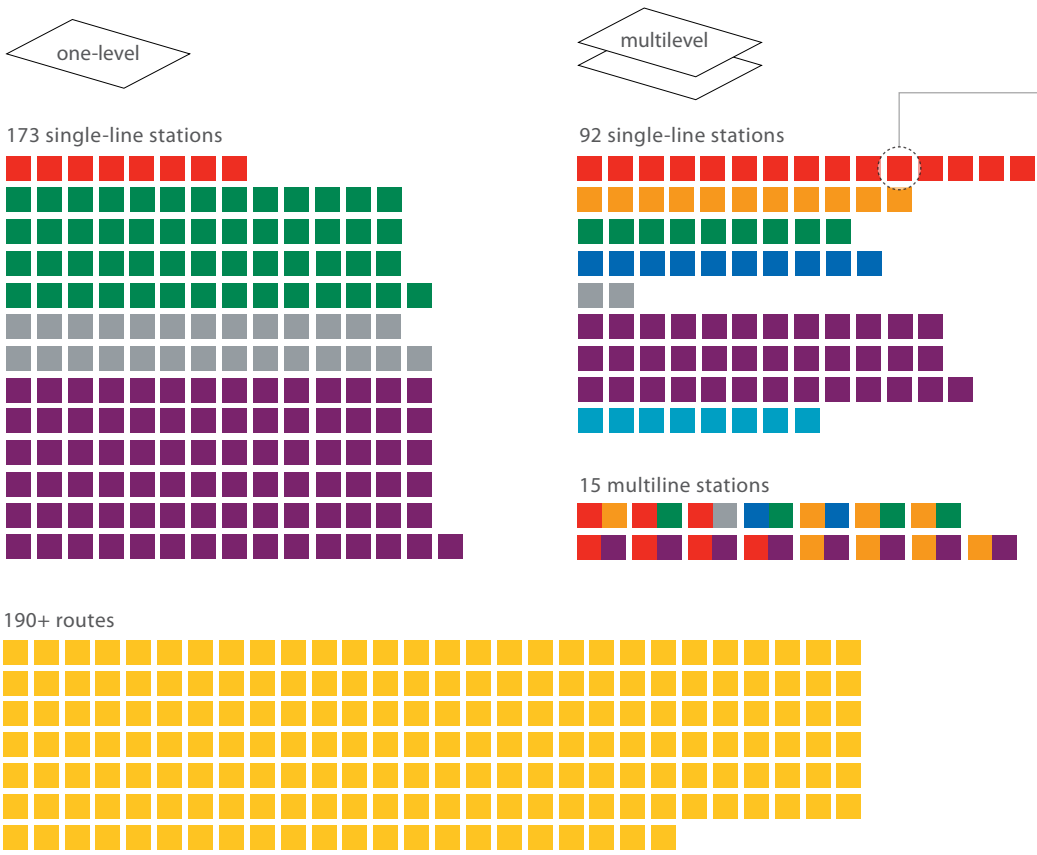

COMMUTER RAIL


Boat


FERRY


Bus


BUS



Website for Each Station

**CENTRAL**

STATION OVERVIEW

STATION	
Name:	CENTRAL
Branch:	Main
Line:	Red Line

PLATFORM	
Type:	Side

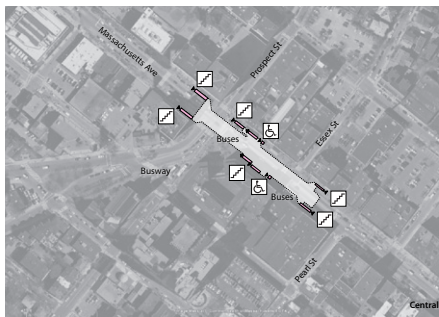
CONNECTIONS	
Parking:	No
Subway:	No
Commuter Rail:	No
Bus:	Yes
Bus Quantity:	8
Bus Configuration:	Single

EXITING	
Exit Configuration:	Direct
Section Type:	2.16

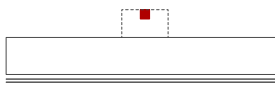
SUPPLEMENTAL	
Emergency Exit:	No
Backlit Signs:	Yes

ACCESSIBILITY	
Accessible:	Yes
Elevator Quantity:	Level 0 : 2
(Platform Config.)	Level -1 : 2 (F2)

EXIT DESTINATIONS	
01	Prospect St
02	Essex St
03	Massachusetts Ave
04	Pearl St
05	City Hall



ELEVATOR PLATFORM CONFIGURATIONS

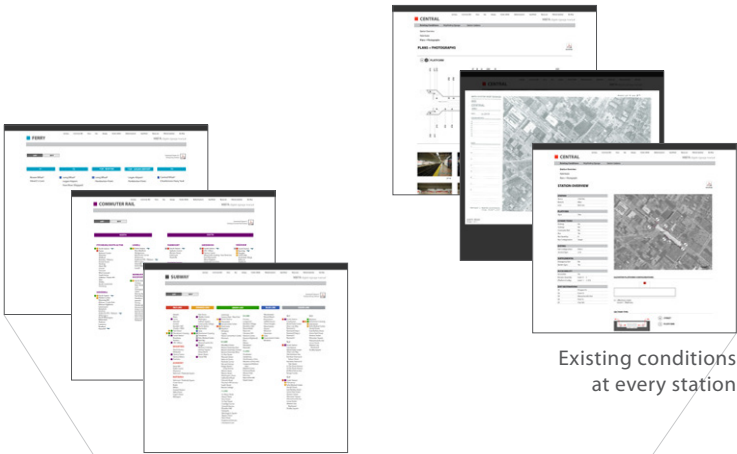


F2 - Offset Out, Center
(Level -1 : Platform)

SECTION TYPE



Section Type 2.16



Existing conditions at every station

All transit modes



DSM Website

Understanding the existing system across all transit modes

The entire MBTA wayfinding signage system is inventoried in BIA's Digital Signage Manual (DSM). The DSM is a vast electronic repository of existing conditions information covering every individual station. The DSM organizes over 30,000 files, including 10,000 photographs, plans and sections of every station, as well as detailed signage layouts and sign artwork for re-signing every station.



The Problem: Inconsistency Over Time

INCONSISTENT BUS SIGNAGE

BUS STREET

BUSWAY

INCONSISTENT ICONS

ARROWS

ISA

VERTICAL CIRCULATION

INCONSISTENT STATION IDS

SINGLE-LINE ENTRANCES

GREEN LINE ALL TRAINS

Helvetica Bold

LAWRENCE STATION

Helvetica Medium

MELROSE HIGHLANDS

Helvetica Condensed

CENTRAL AVE

Microgamma Bold / Helvetica Medium

ALEWIFE

Microgamma Bold / Eurostile Light

MULTILINE ENTRANCES

FIELDS CORNER ALL TRAINS

GREEN TO TRAINS

MAVERICK ALL TRAINS

BRAINTREE

FOREST HILLS ORANGE LINE

MALDEN CENTER

HAYMARKET HAYMARKET TO TRAINS

BOYLSTON STREET INBOUND

COURTHOUSE STATION

SHARON OUTBOUND

COMMUTER RAIL TO TRAINS

PORTER ENTRANCE TO ALL TRAINS

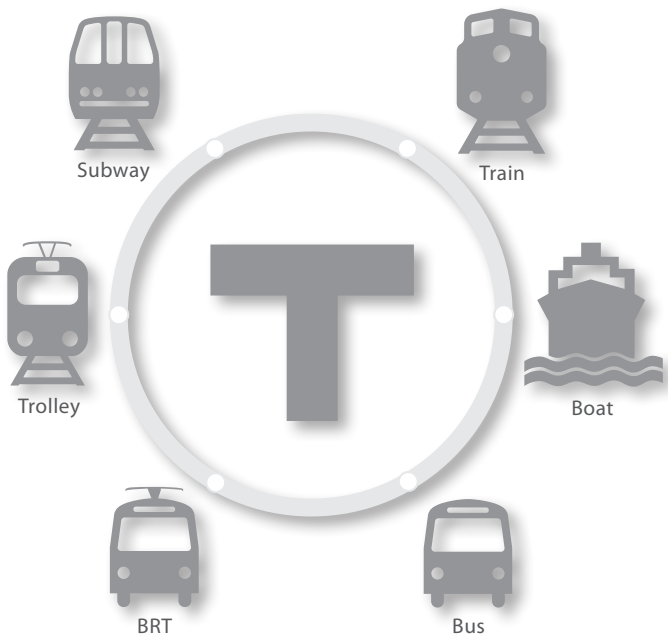
SILVER LINE RED LINE SOUTH STATION

Analyzing the existing system to understand the design challenges

As the oldest transit system in the US, the MBTA has evolved many different ways of displaying wayfinding information and station identification. A complete inventory of every sign at every station establishes a baseline for conducting a thorough diagnosis of the overall system. This is the first step in creating a wayfinding system that is Correct, Clear, and Consistent.

AD HOC SIGNAGE

The Solution: A Unified Signage System



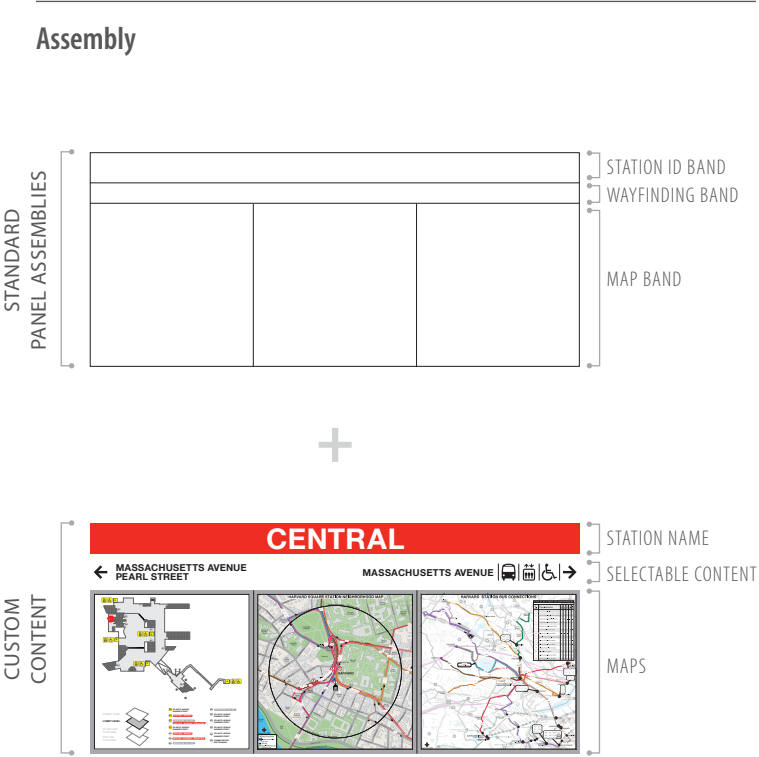
One-System Map

Creating one family of signage

In recent years the MBTA has shifted focus to be more customer-centered, regarding the transit system as one interconnected network. The overarching goal of the new wayfinding system is to create a sense of unity among all modes of transit in order to improve customer perceptions of quality and dependability.

SINGLE-LINE STATIONS	MULTILINE STATIONS	BUS	TACTILE/BRAILLE
HARVARD RED LINE - ALL TRAINS Red Line CHINATOWN ORANGE LINE - OAK GROVE Orange Line AQUARIUM BLUE LINE - ALL TRAINS Blue Line COPLEY GREEN LINE - GOV CTR & NORTH Green Line WORLD TRADE CTR SILVER LINE - SL1/SL2 Silver Line CONCORD OUTBOUND Commuter Rail CENTRAL WHARF F4 - CHARLESTOWN NAVY YARD Ferry	PARK STREET RED LINE - ALL TRAINS GREEN LINE - COPLEY & WEST DOWNTOWN CROSSING ORANGE LINE - ALL TRAINS RED LINE - ALL TRAINS GOVERNMENT CENTER BLUE LINE - ALL TRAINS GREEN LINE - ALL TRAINS NORTH STATION GREEN LINE - ALL TRAINS ORANGE LINE - ALL TRAINS SOUTH STATION SILVER LINE - SL1/SL2 RED LINE - ALL TRAINS COMMUTER RAIL AMTRAK FRAMINGHAM COMMUTER RAIL AMTRAK	BERTH A 51 Forest Hills via Cleveland Circle 111 Washington Ave at Revere Beach Pkwy Busway 51 Forest Hills via Cleveland Circle 111 Washington Ave at Revere Beach Pkwy NO STOPPING TOW ZONE Bus Street	BROADWAY STAIRS TO: EXIT, LOBBY ELEVATOR Directional STATE STAIRS TO: ORANGE LINE - ALL TRAINS, BLUE LINE - ALL TRAINS ACCESSIBLE ENTRY MAP Entry

Signage DNA



Standardizing content elements

The new wayfinding signage system combines an extensive inventory of standard panel assemblies with the ability to apply customized, data-driven content to create location-specific wayfinding signage. A library of predefined colors, path icons, symbols, maps and fonts form a common vocabulary of communication for all modes of transit.

Identity

COLOR BANDS

- Red Line
- Orange Line
- Green Line
- Blue Line
- Silver Line
- Commuter Rail
- Ferry
- Bus

T LOGO

Maps

ENTERING

EXITING

System

Line

Busway

Neighborhood

Neighborhood Bus

Icons

PATHS

SYMBOLS

Ambulatory

Accessible

Destinations

Custom-kerned fonts create consistent, ADA-compliant text:

HELVETICA NEUE MBTA BOLD

Helvetica Neue ADA Bus

LINE - DESTINATION

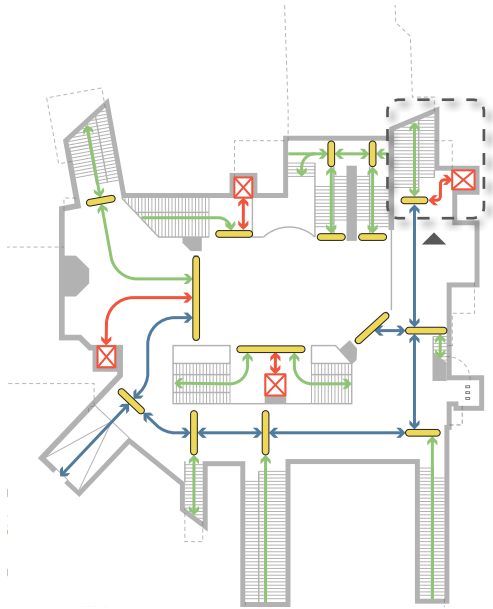
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Pathway Clarity

PATHWAY DIAGRAM

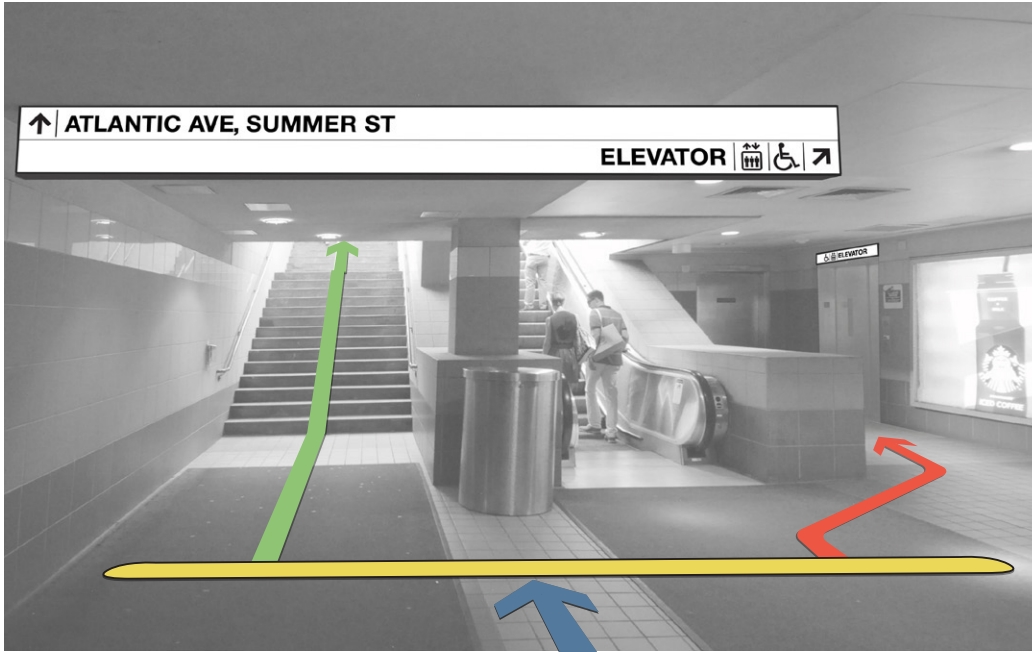
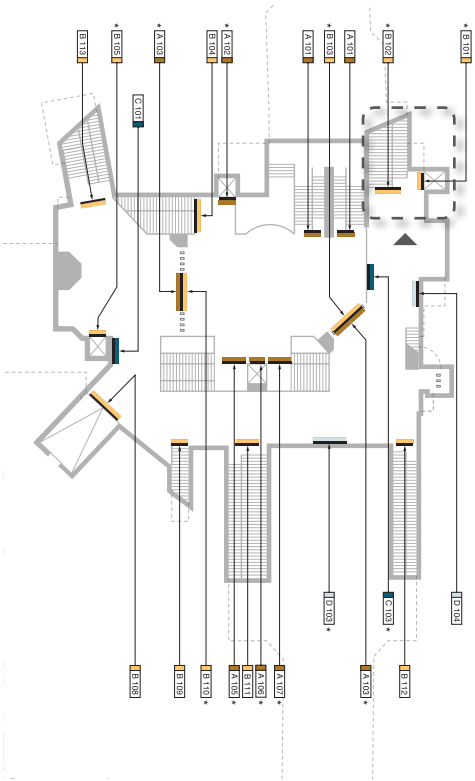


Decision point:
ambulatory & accessible paths split



- Accessible
- Ambulatory
- Combined
- Decision Point

SIGN LAYOUT

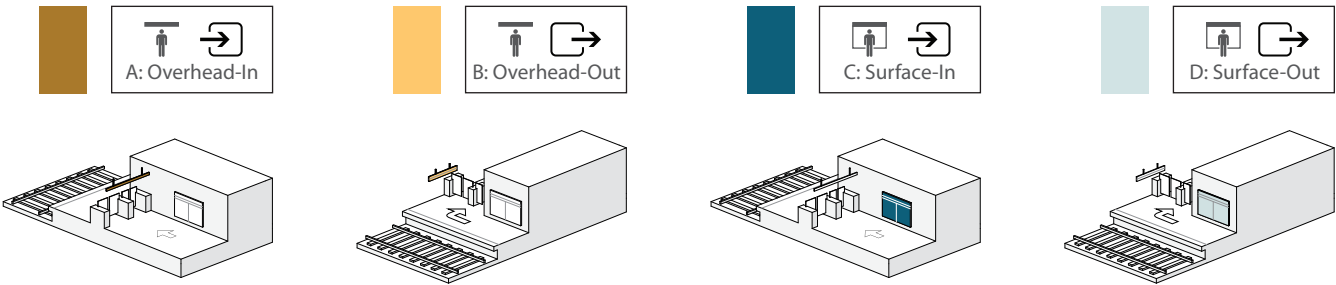


Signed decision point:
ambulatory & accessible paths split

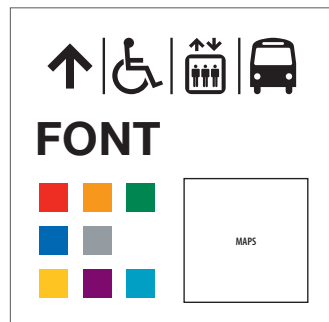
Locating decision points

Decision points are places along a circulation route where a path splits between accessible (to elevators or ramps) and ambulatory (stairs or escalators), or where paths to different destinations diverge. The new wayfinding system is based on locating each and every path split within a station and placing signage to help customers continue along the appropriate path to their desired destination.

4 Sign Types

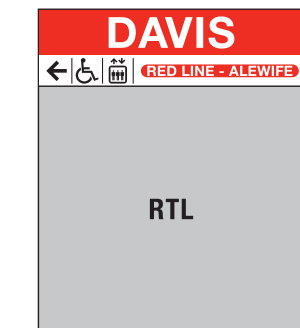


Keeping it Simple: CCC



Correct

In order for a wayfinding system to work correctly, content must be accurate, code-compliant, and organized on signs in standard arrangements. Correct information and display is the foundation of a successful signage system.



Clear

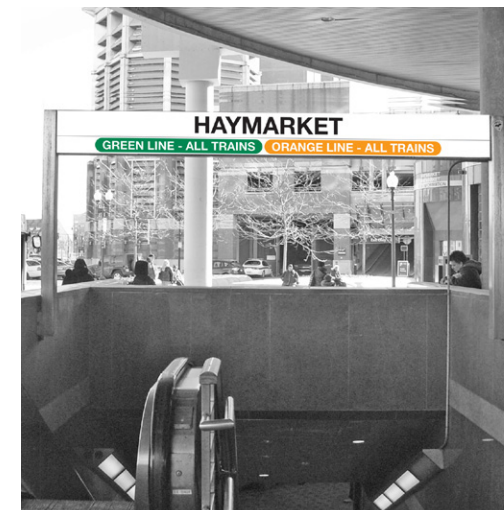
This is the quality that makes correct information intelligible to customers – the judicious display of wayfinding information, arranged and presented to maximize the customer’s ability to successfully navigate individual stations and the system.



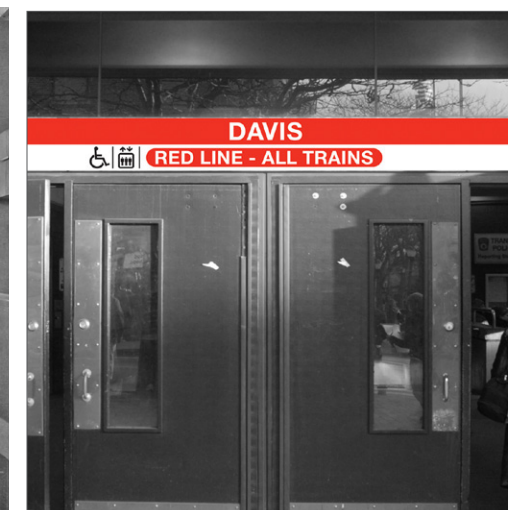
Consistent

The purpose of consistency is to instill trust in the users of a system – this is the sense that each station uses the same methods and conventions of communication. In turn, customers develop confidence that the signage everywhere is accurate and reliable.

Subway multiline stations



Subway single-line stations



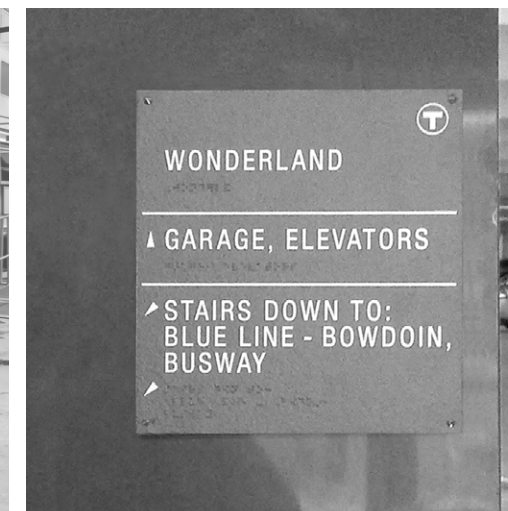
Commuter Rail



Ferry



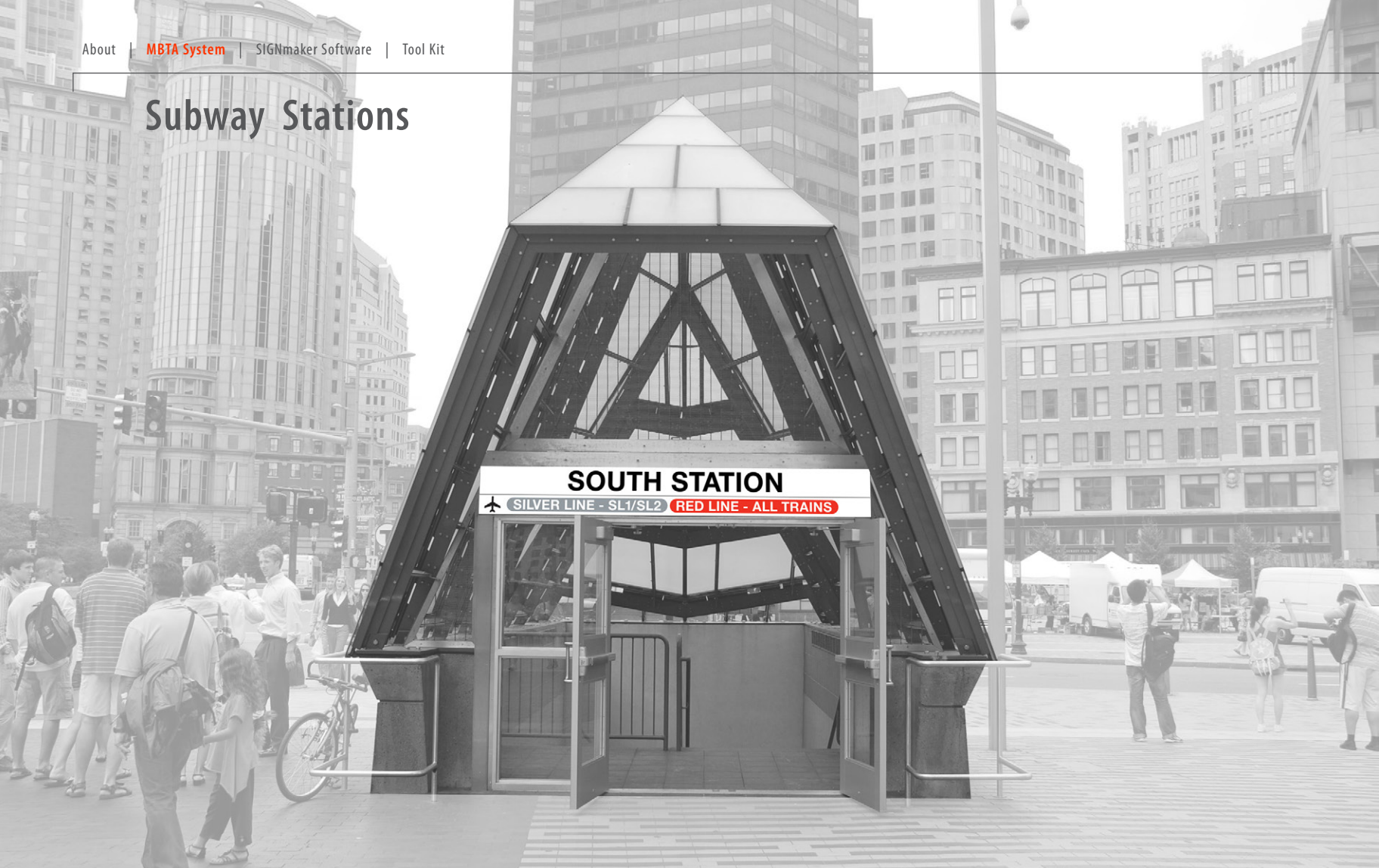
Tactile/Braille



Bus



Subway Stations



Using color to enhance wayfinding

Within the MBTA system, subway lines are differentiated and named by color (Red Line, Blue Line, etc). The new wayfinding system uses color ovals to display each line and destination, clarifying which lines and destinations are available within a station. For customers, this greatly improves the wayfinding information hierarchy along circulation paths, especially in complex multiline, multilevel stations.

entering



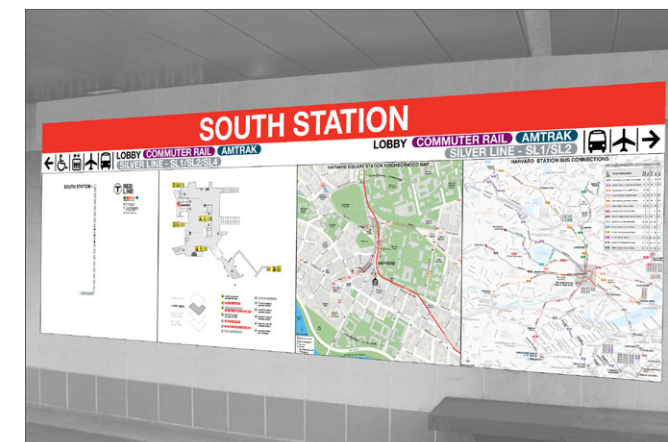
overhead



entering
↓



exiting
↑



Application of the 4 sign types at a multiline subway station

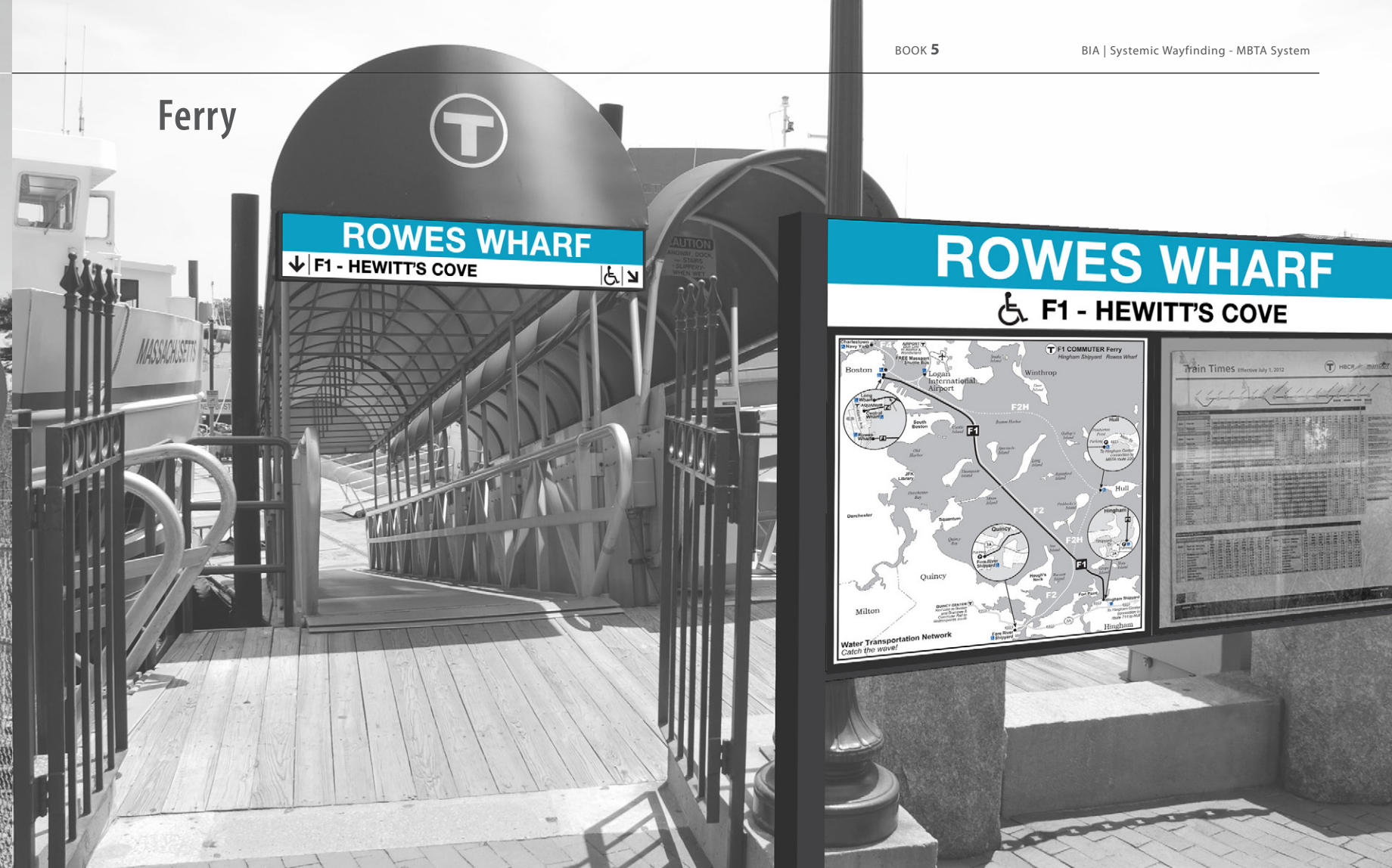
Commuter Rail



Complementary rail signage

The MBTA commuter rail system consists of 10 lines and 130 stations. All lines are brand-identified with the commuter rail's signature purple color. Wayfinding signs draw on the same library of graphic components and conventions as used on the subway system, adapted to serve the signage needs of the commuter rail.

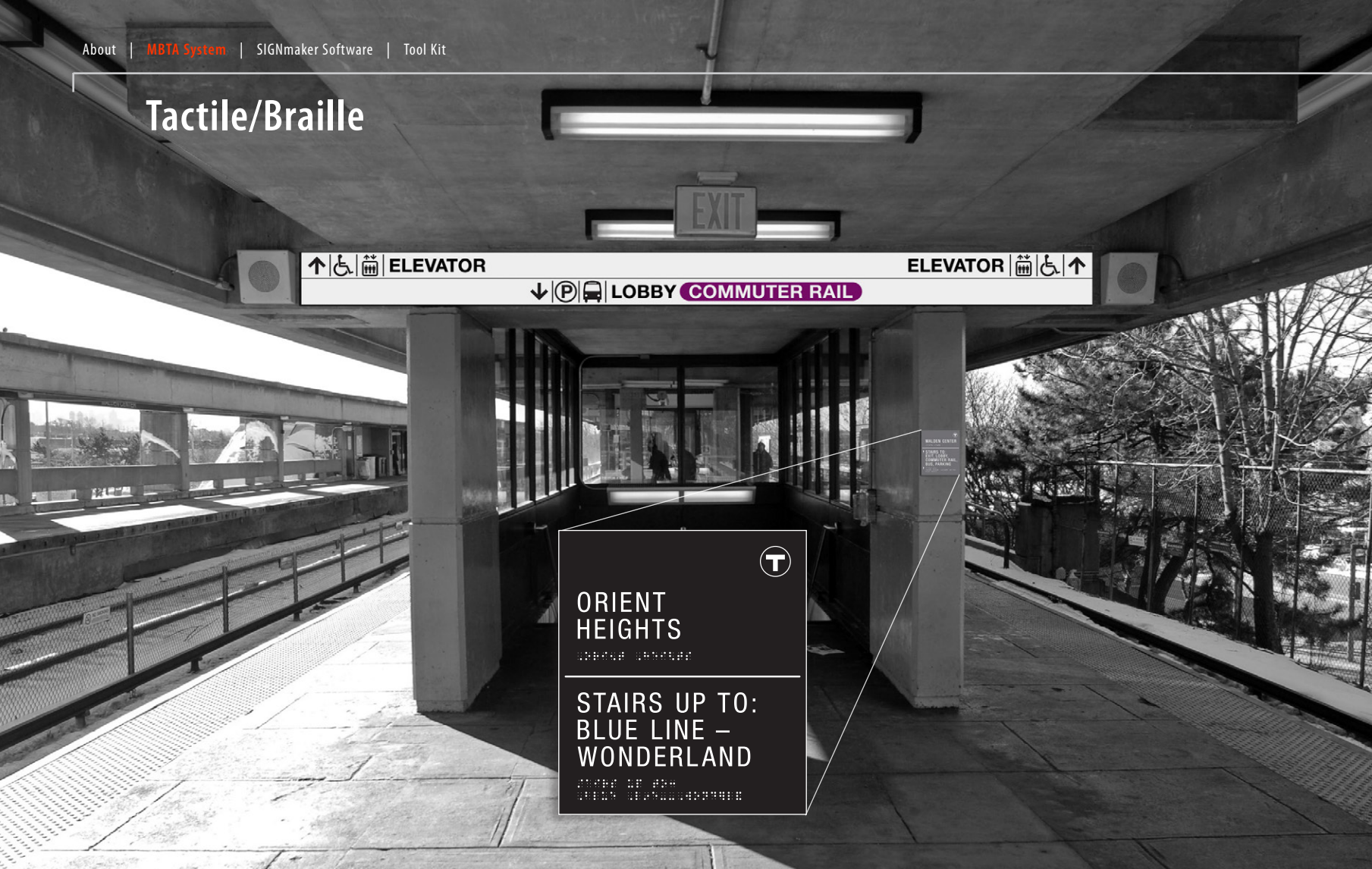
Ferry



Incorporating the ferry system

The MBTA operates 4 ferry routes between 8 ferry terminals in Boston Harbor. The ferry system is color coded with light blue to signify another “line” of transit, using the subway’s traditional color banding system to identify terminals as part of the unified MBTA system. Wayfinding signage is typically simple, focusing on identifying destinations and locations of accessible paths.

Tactile/Braille



ADA-compliant signage

Tactile/Braille signs are located at specific decision points within a station, including station entries and platforms, as well as at vertical circulation. The content of each sign parallels the adjacent visual wayfinding signage. The ADA-compliant graphic layout is automatically generated from menu selections made in SIGNmaker™ TB.

Bus



Bus street and busway signage

There are over 8,100 bus stops and 28 busways on the MBTA system. The new wayfinding design uses the color yellow to brand the bus system. Standard templates were created for both freestanding street signs and bus berths at station busways. The content and graphic layout of each sign is automatically generated from Bus Operation's database of routes and stops within SIGNmaker™ BUS.



SIGNmaker™

Export: generate SIGNmaker™ Output

Station: identify selected station

Sign: identify sign tag / create signs

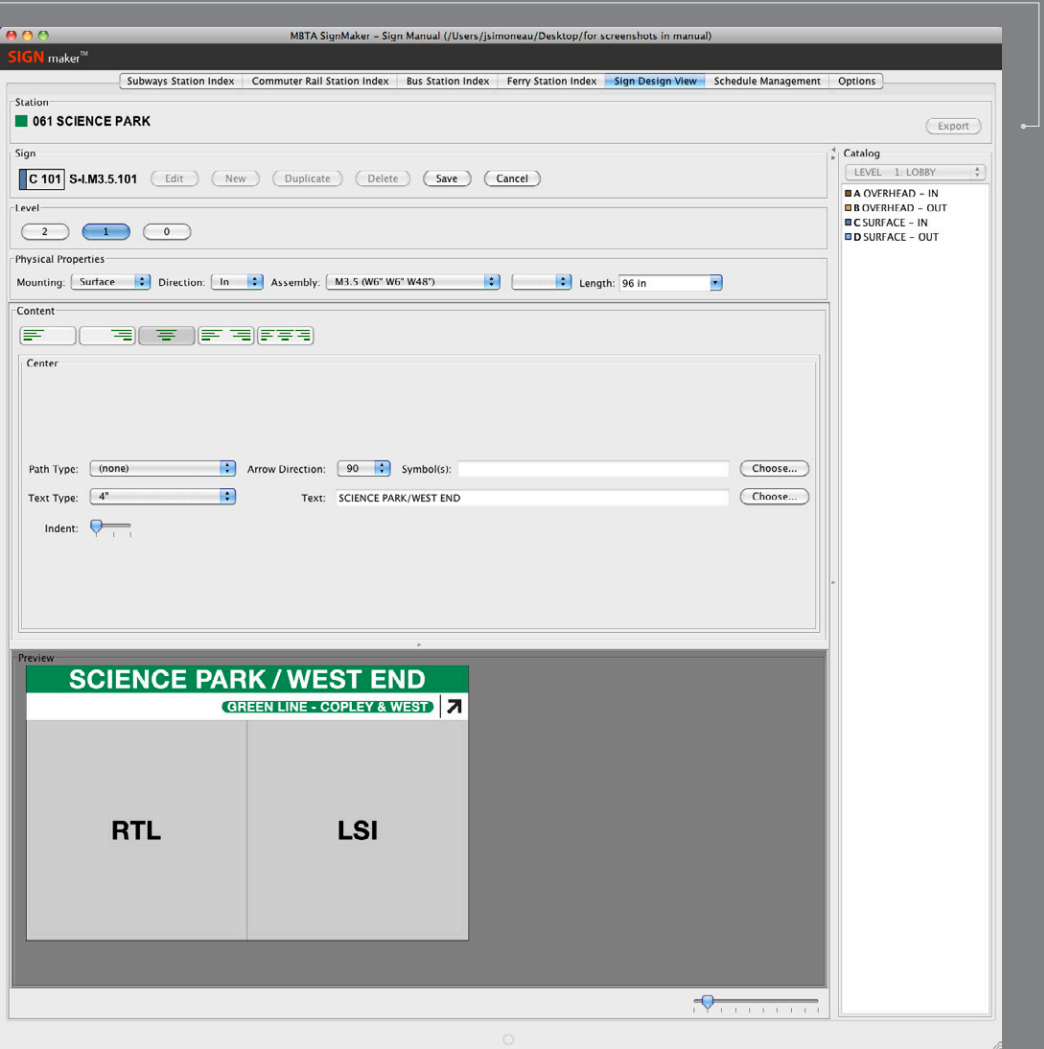
Level: select station level

Physical Properties: select sign type,
panel assembly, size

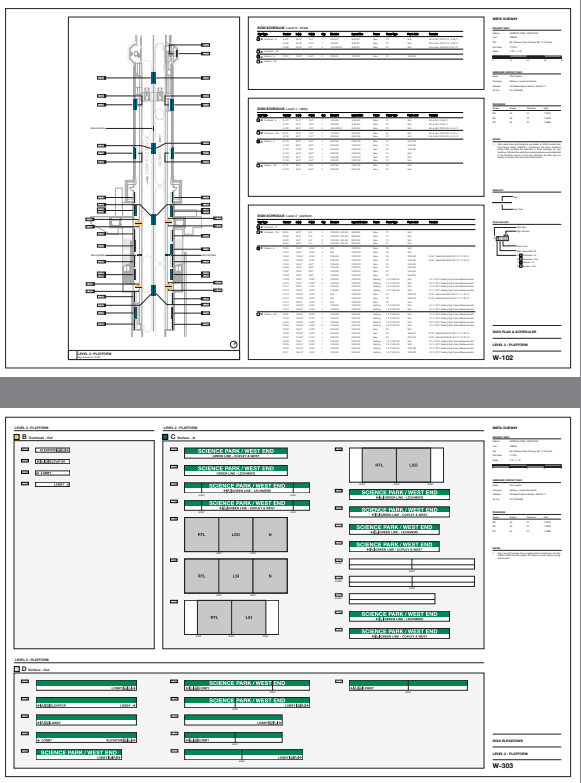
Content Selection: select panel justification
/ add paths, text, icons / adjust text size,
arrow direction, content alignment

Sign Preview: review signs

Sign Catalog: view created signs
by station level



Create



Construction Documents



Produce

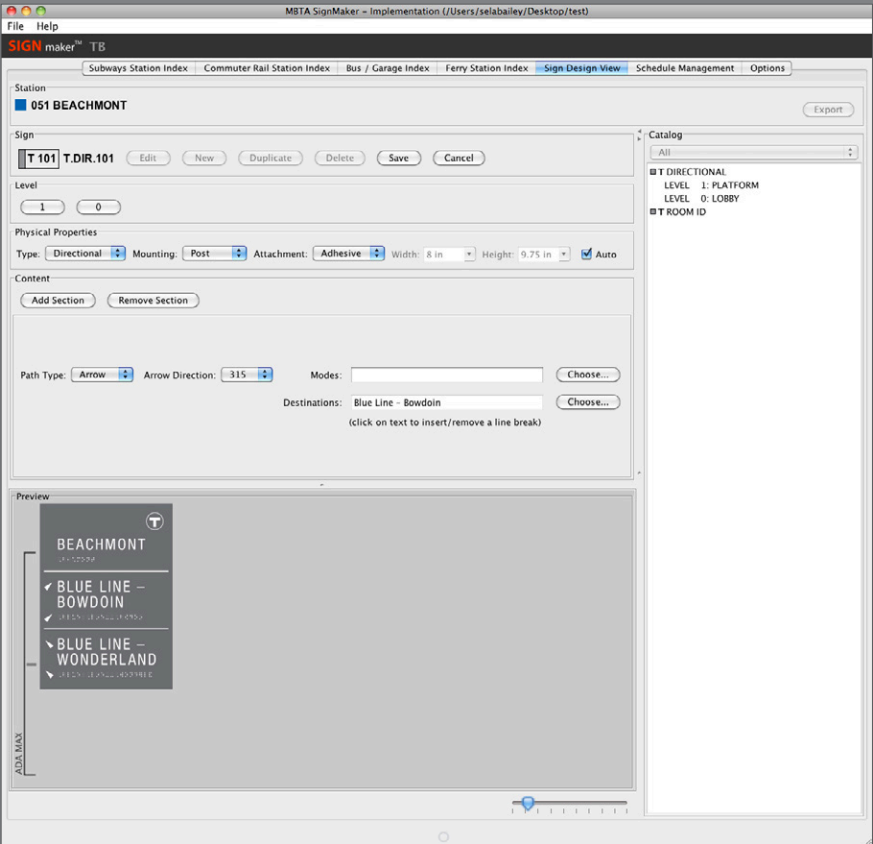


Finished Product

Automates the creation of correct,
consistent signage

SIGNmaker™ is an innovative, menu-driven software application custom-developed for automating the design process. The interface automatically positions all signage elements, including sign panel sizes, colors, text, symbols, and maps. SIGNmaker™ simplifies the process of designing Subway, Commuter Rail and Ferry signage, and ensures that signs are correct, clear, and consistent from station to station.

SIGNmaker™ TB

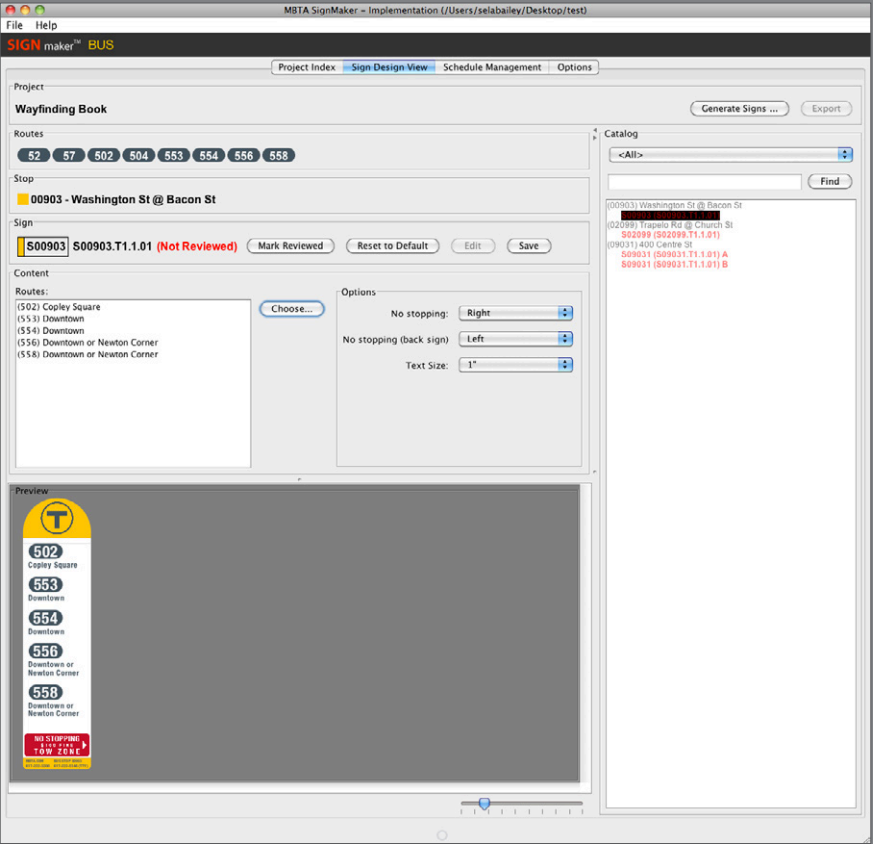


Generate

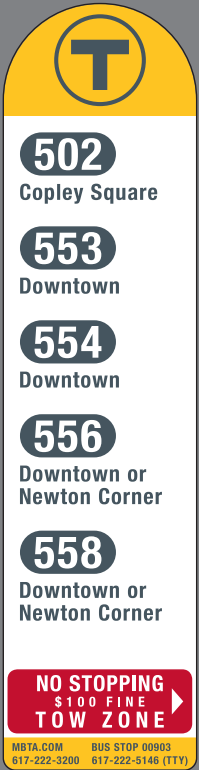


Etched zinc tactile/braille sign

SIGNmaker™ BUS



Generate



Painted retro-reflective aluminum sign

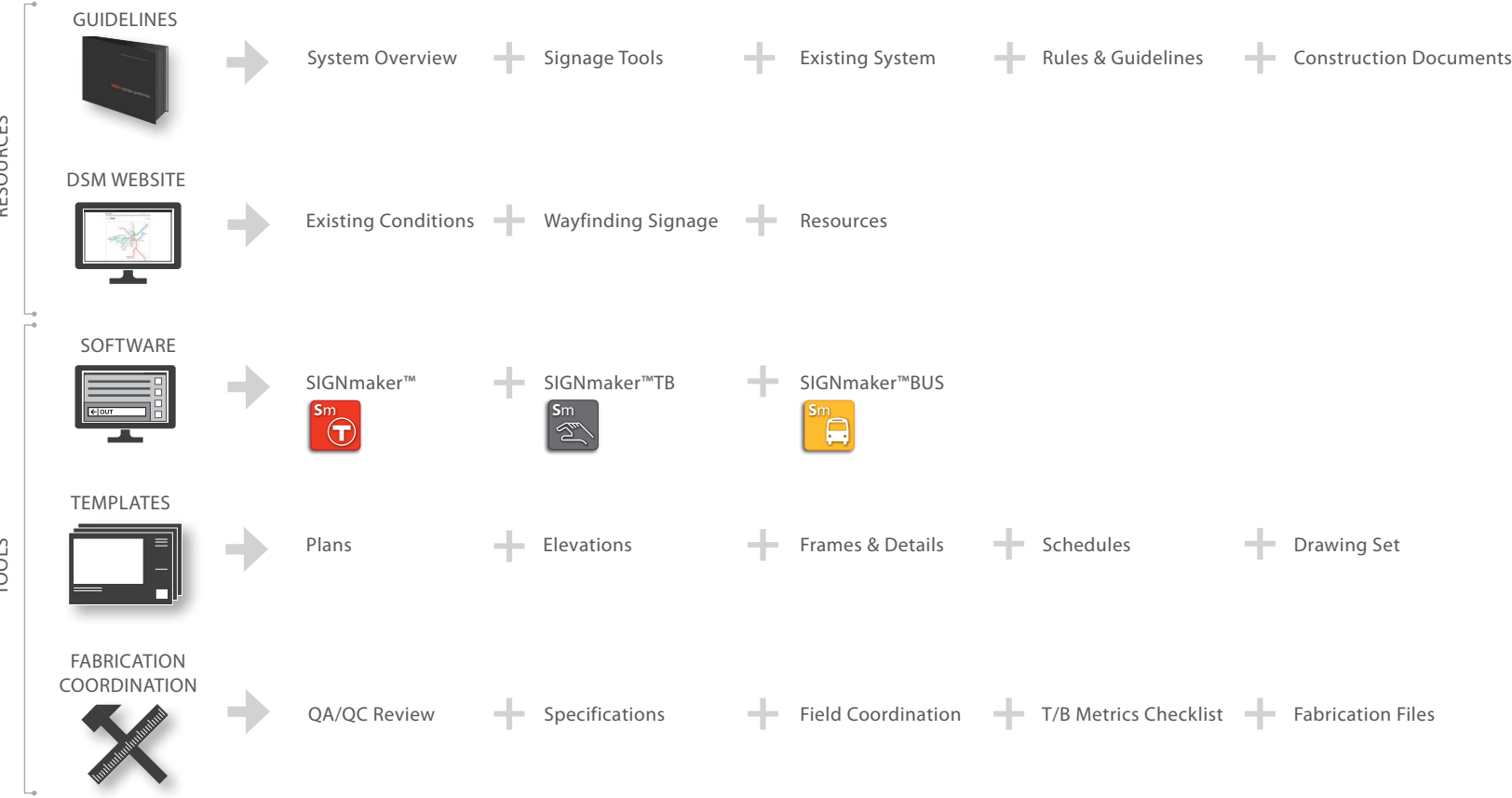
Customizable software

The SIGNmaker™ TB interface enables automatic generation of full size, ADA-compliant Tactile/Braille signs. Each sign is based on a standard template that can be manipulated to fit content. Each sign displays the station name with directional information modules inserted below. The software can be easily customized to include an identifying system logo or title on each sign.

Simplified production

The SIGNmaker™ BUS interface enables automatic generation of full size bus stop signs on two standard template sizes. The software is linked to Bus Operation's route database and serves as a content management tool, facilitating rapid generation of street signs for an entire route without any need for time-consuming manual design.

Comprehensive Wayfinding Solution



Systematic approach to in-house production

SIGNmaker Wayfinding offers a complete suite of tools to meet every wayfinding design challenge. Leveraging the power of data and system-wide graphic conventions, this system offers an unparalleled custom approach for in-house execution of every aspect of signage design and fabrication.





118 South Street • Boston • MA 02111 **T:** (617) 423-6500 **W:** bia.studio